

PREDICTION TASK ? The task is to classify Steam game reviews as either positive or negative based on the text content of the reviews. The input is the review text, and the output is a binary sentiment label (positive or negative). The goal is to automatically predict the sentiment of new reviews to help Steam and game developers understand user feedback more efficiently.	PREDICTION TASTK ↕ The sentiment analysis model serves as a tool for Steam and game developers to automatically classify user reviews. This helps in quickly identifying positive and negative feedback, allowing developers to prioritize improvements and Steam to better recommend games to users. The model can also be used to analyze reviews from external forums, providing a broader understanding of game popularity and user satisfaction. Parameter: Review text, user recommendations, and other metadata (e.g., game title, user ID).	VALUE PROPOSITION 🎁 - User Target: Steam platform, game developers, and marketers. - User Objective: Automatically classify user reviews to understand sentiment and improve decision-making. - User Benefit: Saves time by automating sentiment analysis, provides actionable insights from user feedback, and enhances user experience by identifying popular and problematic games.	DATA COLLECTION ⬇️ The dataset consists of Steam reviews for 64 game titles, including the review text and whether the user recommended the game. The data is preprocessed to remove noise (e.g., special characters, stopwords) and prepared for machine learning models. If the model's performance is unsatisfactory, it can be retrained with additional reviews or fine-tuned with hyperparameter optimization.	DATA SOURCES 🗄️ - Primary Data Source: Steam reviews dataset, which includes review text, user recommendations, and game-related metadata. - Secondary Data Source: External forums or review platforms (if needed for additional training data).
IMPACT SIMULATION ✓ The model's performance is evaluated using metrics such as accuracy, precision, recall, F1-score, and confusion matrices. These metrics help determine how well the model classifies reviews into positive and negative sentiments. Additionally, visual tools like WordCloud and sentiment distribution analysis are used to interpret the results and validate the model's predictions. - Custom Metric: The F1-score is particularly important for imbalanced datasets, ensuring that both positive and negative reviews are classified accurately. - Visual Validation: WordClouds and sentiment distribution plots help in understanding the most frequent words and overall sentiment trends.	MAKING PREDICTIONS ↔️ The model will classify reviews in batches rather than in real-time. This means that a large number of reviews will be processed at once, and the results will be stored in a database for later use. - Steam reviews are typically submitted in large volumes, and real-time processing is not necessary for sentiment analysis. - Batch processing is more efficient for large datasets and reduces computational overhead.	BUILDING MODELS ⚙️ The project implements multiple models for sentiment analysis: - Traditional Machine Learning Models: - Naive Bayes: Efficient for text classification, especially with TF-IDF vectors. - Logistic Regression: Provides a good balance between accuracy and training time. - Deep Learning Model: - LSTM (Long Short-Term Memory): Captures sequential dependencies in text, suitable for understanding context in reviews. - The models are trained using TF-IDF for feature extraction and GloVe embeddings for the LSTM model.		FEATURES 📊 - Text Data - Review Text: The primary input for sentiment analysis. - Metadata - User Recommendations: Binary data indicating whether the user recommended the game (1 for recommended, 0 for not recommended). - Game Title: The title of the game being reviewed. - User ID: Unique identifier for the user who wrote the review. - Feature Extraction - TF-IDF (Term Frequency-Inverse Document Frequency): Captures the importance of words in the context of the entire dataset. - Word Embeddings: - Pre-trained GloVe embeddings are loaded to provide semantic meaning for each word as a vector. - Additional Features - Sentiment Scores: - analysis library like VADER, sentiment scores for individual words or phrases can be included as features. - Word Frequency: The frequency of specific words in positive and negative reviews used as additional features.
MONITORING 🔄 To monitor the model's performance, periodic evaluations are conducted using the test set and new reviews. Feedback from Steam and game developers can be collected to assess the model's real-world effectiveness. If the model's performance degrades over time (e.g., due to changes in review patterns), it can be retrained with updated data. - Feedback Mechanism: Developers can provide feedback on the accuracy of sentiment predictions, which can be used to fine-tune the model. - Retraining: The model can be retrained periodically to adapt to new review trends and maintain high accuracy.				